Time-Varying Nonlinear Factor Models via Deep Autoencoders

Preliminary draft

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1 Introduction

Factor models are a workhorse of modern empirical economics. In macroeconomics they underpin the diffusion index forecasts of Stock and Watson (1999) and Stock and Watson (2002); in finance they are the empirical foundation of the arbitrage pricing theory of Ross (1976) and its empirical descendants (Fama and French, 1993, 2015); in panel econometrics they provide a flexible device for controlling unobserved heterogeneity (Bai, 2009; Pesaran, 2006). In all these settings the same statistical structure is assumed: the cross section of N observed variables is driven by a small number of latent factors $F_t \in \mathbb{R}^K$, with each unit's exposure specified as a linear map from the factors to the observation: $X_{it} = \lambda_i' F_t + U_{it}$.

Two features of this structure have come under sustained scrutiny in recent years. First, the linearity of the loading function is a convenience rather than a theoretical requirement. Almost every structural model in finance and macroeconomics predicts nonlinear dependence of individual returns or outcomes on aggregate state variables (Bansal and Yaron, 2004; Campbell and Cochrane, 1999; Santos and Veronesi, 2004). In practice, the linear approximation can be inaccurate in ways that matter for pricing and for inference. Shen and Xiu (2026) formalize this concern and propose a deep autoencoder estimator that recovers nonlinear loading functions with non-asymptotic guarantees. Second, the assumption that loadings are fixed over a long horizon has been challenged by evidence of structural change and regime shifts in financial markets (Breitung and Eickmeier, 2011; Stock and Watson, 2009). L. Su and Wang (2017) formalize this concern and propose a local principal component estimator that recovers time-varying linear loadings, again with asymptotic guarantees.

Both extensions are natural. They have not previously been combined. Shen and Xiu (2026) assume constant loadings; L. Su and Wang (2017) assume linearity. In applications to equity returns, portfolio construction, and macro-financial forecasting, both assumptions fail simultaneously: loading functions are nonlinear in the factors and the nonlinearity shifts with the macroeconomic regime. The natural specification is therefore

$$X_{it} = \varphi_i^\star(F_t^\star,\, t/T) + U_{it}, \tag{1}$$

where $\varphi_i^\star(\cdot,\cdot)$ is an unknown smooth function of the K-dimensional latent factor $F_t^\star$ and the scaled time index t/T.

This paper develops a theory and an estimator for model (1), using deep autoencoders. An autoencoder is a neural network trained to reconstruct its own input through a low-dimensional bottleneck layer. When the bottleneck has K_1 neurons, the network provides a nonlinear analogue of principal components analysis: the bottleneck acts as a set of latent factors, and the decoder acts as a set of loading functions. When the bottleneck is further conditioned on the scaled time index t/T, the decoder becomes a time-varying loading function. This construction gives a natural estimator for (1) and connects the nonlinear factor structure of Shen and Xiu (2026) to the time-varying factor structure of L. Su and Wang (2017).

Two natural candidates for the estimator differ in how they incorporate time. The *local autoencoder* (TV-AE-L) is the direct nonlinear analogue of L. Su and Wang (2017): at each of a grid of reference times, a separate autoencoder is trained by minimizing a kernel-weighted reconstruction loss. The *global time-conditioned autoencoder* (TV-AE-G) replaces the grid with a single autoencoder that takes the scaled time index as an input to both the encoder and the decoder. We show that TV-AE-G attains a strictly faster rate than TV-AE-L: the local estimator pays a kernel-smoothing penalty, while the global estimator exploits the smoothness of the loading function in t/T without a bandwidth. The rate improvement is a central theoretical contribution of the paper. A third variant, the *characteristic-conditioned autoencoder* (TV-AE-C), extends TV-AE-G to also condition the decoder on observable unit characteristics, following the conditional autoencoder architecture of Gu, B. Kelly, and Xiu (2021).

Our theoretical contributions are three. First, we establish a non-asymptotic bound for TV-AE-G. The rate is $T^{-2\beta/(2\beta+K+1)} + N^{-1}K_1 + \epsilon_N$, where the first term is the nonparametric rate for estimating the loading function $\varphi_i^\star(F, s)$ in $K+1$ effective dimensions, the second is the parametric estimation error for the TK_1 unknown factor values, and the third is the encoder approximation error. The rate strictly dominates the rate of the local autoencoder analogue of L. Su and Wang (2017), and it reduces to the Shen and Xiu (2026) rate $T^{-2\beta/(2\beta+K)}$ when the loading function is constant in t. Second, we extend the result to TV-AE-C, where the rate becomes $T^{-2\beta/(2\beta+K+P+1)} + N^{-1}K_1 + \epsilon_N$, quantifying the cost of conditioning on P observable characteristics. Third, we show that the estimated factors are identified up to a time-varying invertible transformation, and we discuss the identification of the number of factors in a nonlinear setting.

Our simulation studies evaluate these predictions on a grid of data-generating processes that vary the form of the nonlinearity (linear, exponential, quadratic) and the form of the time variation (constant, logistic). The results align with the theory. When the loadings are linear and time-invariant, TV-AE-G ties the static autoencoder of Shen and Xiu (2026). When the loadings are nonlinear and time-varying, TV-AE-G outperforms both the static autoencoder and the local principal component estimator of L. Su and Wang (2017) by substantial margins (20–35% in mean squared error at the sample sizes we consider). When the dominant nonlinearity is time-invariant but the time variation is in a smaller component, the static autoencoder can overtake TV-AE-G at sufficiently large T. This last pattern is not a failure of the estimator: it reflects a genuine bias-variance tradeoff, and it is exactly what the theory predicts.

Our empirical section uses a four-panel design that isolates the conditions under which the autoencoder's nonlinearity provides value. The four panels are (i) individual equity returns

on the S&P 100, (ii) individual equity realized volatilities, (iii) a panel of crypto-currency daily returns, and (iv) crypto-currency realized volatilities. In each panel we compare the out-of-sample reconstruction R^2 of static PCA, the static autoencoder, and TV-AE-G, using a rolling estimation window with monthly refits. The pattern is consistent with the theory and with the simulation evidence. On the equity-return and equity-volatility panels, the autoencoder matches PCA but does not beat it: the linear factor structure is not misspecified at these sample sizes. On the crypto-volatility panel the same holds: both methods achieve R^2 above 0.97 and the difference is not statistically distinguishable from zero. On the crypto-return panel, by contrast, TV-AE-G beats static PCA by 3.7 percentage points, and a paired bootstrap over refit windows excludes zero at the 5% level.

We also examine a downstream application to minimum variance portfolio construction on the S&P 100. On this panel the autoencoder does not outperform linear shrinkage. We interpret this as a genuine finding rather than a limitation: minimum variance portfolios are dominated by the smallest eigenvalues of the covariance matrix, and the autoencoder's small-sample noise in those directions outweighs any gain in the factor structure.

Finally, Section 6.4 documents two applications that we attempted and did not pursue to a successful conclusion: an equity-return prediction exercise, in which the encoder collapsed to the per-stock mean because the monthly signal-to-noise ratio is too low for gradient descent; and a set of local-autoencoder and local-PCA runs at short estimation windows, in which the effective sample per grid point was too small for the local methods to train. These failures are reported because they are informative about what the method can and cannot do at the sample sizes we have access to, and because they bear directly on the choice between TV-AE-G and TV-AE-L.

The paper proceeds as follows. Section 3 introduces the model and the autoencoder architectures. Section 4 develops the non-asymptotic theory. Section 5 reports the simulation studies. Section 6 presents the empirical applications. Section 7 concludes. Proofs are collected in the appendices.

Roadmap and what is not yet done. Sections 3–4 are complete except for minor edits to the proof of Proposition 2, which is currently a sketch. Section 5 reports the simulation evidence; the capacity sweep that selected the final hyperparameters is summarized in Section 5.6 but the full sweep is in a separate document. Section 6 reports the empirical work in three parts: the primary reconstruction application (Section 6.2), the secondary minimum-variance application (Section 6.3), and what did not work (Section 6.4). The extensions we consider most urgent are a proper time-extrapolation scheme for the test-time encoder (Section 6.4), a richer characteristic set for TV-AE-C (Section 6.4), and the completion of the factor-recovery proof.

2 Related Literature

Our work sits at the intersection of four literatures.

Linear factor models and their estimation. The modern treatment of large-dimensional factor models originates with Chamberlain and Rothschild (1983) and Connor and Korajczyk (1986), who formalize the approximate factor structure underlying the arbitrage pricing the-

ory of Ross (1976). The asymptotic theory for principal component estimation in large N, large T panels is developed by Stock and Watson (2002), Bai and Ng (2002), and Bai (2003), who establish consistency and asymptotic normality of the estimated factors and loadings under appropriate pervasiveness conditions. This literature assumes loadings that are linear in the factors and constant over time. We extend both assumptions.

Time-varying factor models. Stock and Watson (2009) show that small instabilities in loadings leave PCA factor estimates consistent, but larger instabilities can lead to substantial misestimation. Breitung and Eickmeier (2011) and Chen, Dolado, and Gonzalo (2014) propose tests for one-time structural breaks in factor loadings. L. Su and Wang (2017) introduce the local principal component estimator: loadings are modelled as smooth functions of the scaled time index, and the estimator is a kernel-weighted eigenvalue decomposition at each reference time. Their estimator achieves $\sqrt{N}$ consistency for the factors and $\sqrt{Th}$ consistency for the loadings, with explicit asymptotic distributions. The local estimator pays a kernel-smoothing penalty: the nonparametric rate for the loading function is $(Th)^{-2\beta/(2\beta+K+1)}$ at each reference time, and the effective sample size per fit is only Th. The global time-conditioned autoencoder we propose avoids this penalty by parametrizing the time variation through a neural network input rather than kernel weighting. The rate improves from $T^{-2\beta/(2\beta+K+2)}$ (the local analogue) to $T^{-2\beta/(2\beta+K+1)}$ (Theorem 1 below), a gap that widens with K and β.

Nonlinear factor models and autoencoders. The nonlinear analogue of PCA via neural networks has a long history in the machine learning literature, beginning with the work of Baldi and Hornik (1989) on linear autoencoders and Hinton and Salakhutdinov (2006) on deep autoencoders for image data. Gu, B. Kelly, and Xiu (2021) introduce a conditional autoencoder that allows factor exposures to depend on observable characteristics, building on the instrumented principal component model of B. T. Kelly, Pruitt, and Y. Su (2019). Shen and Xiu (2026) provide the non-asymptotic theoretical foundation for autoencoder factor models with constant loadings, establishing rates that match nonparametric minimax bounds up to logarithmic factors and that reduce to the linear factor model rates as a special case. Our work extends Shen and Xiu (2026) by adding time variation in the loading function.

Deep learning theory for nonparametric estimation. The approximation and estimation properties of deep ReLU networks in nonparametric regression have been studied extensively over the past decade. Schmidt-Hieber (2020) establishes minimax optimality of deep ReLU networks for Hölder-smooth functions under sparsity assumptions on the network weights. Kohler and Langer (2021) removes the sparsity assumption at the cost of a slightly weaker rate, and this is the approximation lemma we use. Farrell, Liang, and Misra (2021) provide estimation and inference results for deep neural networks in a general nonparametric setting. Jiao et al. (2023) establish non-asymptotic error bounds for deep nonparametric regression on approximate manifolds.

3 Model

3.1 A time-varying nonlinear factor model

Let $\{X_{it}\}$ denote a panel indexed by $i = 1, \ldots, N$ units and $t = 1, \ldots, T$ time periods. Following the nonlinear factor framework introduced by Amemiya and Yalcin (2001), extended to accommodate time-varying factor loadings, the observed data are assumed to admit the decomposition

$$X_{it} \;=\; X^{\star}_{it} + U_{it} \;=\; \varphi^{\star}_i(F^{\star}_t,\, t/T) + U_{it}, \qquad i = 1, \ldots, N, \quad t = 1, \ldots, T, \tag{2}$$

where $X^{\star}_{it}$ denotes the common component, $\varphi^{\star}_i(\cdot, \cdot) : \mathbb{R}^K \times [0, 1] \to \mathbb{R}$ is the loading function of unit i, $F^{\star}_t \in \mathbb{R}^K$ is a K-dimensional vector of latent factors, and U_{it} is an idiosyncratic error term.

Two features of (2) distinguish it from the factor models that dominate empirical work. The first is that $\varphi^{\star}_i(\cdot, s)$ is an unknown function rather than a linear map from the factors to the observation. Linearity of the loading function is a convenience rather than a theoretical requirement: most structural models in finance and macroeconomics predict nonlinear dependence of individual outcomes on aggregate state variables, and Shen and Xiu (2026) show that when the linear approximation fails, a deep autoencoder recovers the loading functions at the optimal nonparametric rate. The second is that the loading function is allowed to change over time. L. Su and Wang (2017) argue that constancy of loadings over a long horizon is untenable when the driving forces of structural change—institutional switching, preference changes, technological progress—operate gradually, and they propose a local principal component estimator for loadings that vary smoothly in t/T.

The two features are logically independent. Shen and Xiu (2026) allow $\varphi^{\star}_i$ to be nonlinear but hold it constant in t; L. Su and Wang (2017) allow $\varphi^{\star}_i$ to vary in t but restrict it to be linear in F. Neither model class nests the other; both are special cases of (2). It is their combination that is the object of the present paper.

Throughout, $X_t = (X_{1t}, \ldots, X_{Nt})^{\top} \in \mathbb{R}^N$ denotes the cross section at time t, $X = (X_1, \ldots, X_T) \in \mathbb{R}^{N \times T}$ the full panel, and $X^{\star}$ and U the corresponding matrices of common components and idiosyncratic errors. The estimand is the common component $X^{\star}$. The factors $F^{\star}_t$ and the loading functions $\varphi^{\star}_i(\cdot, \cdot)$ are not separately identified—for any time-indexed family of invertible transformations μ_t, the reparameterization $F_t \mapsto \mu_t(F_t)$, $\varphi^{\star}_i(\cdot, s) \mapsto \varphi^{\star}_i(\mu_t^{-1}(\cdot), s)$ leaves $X^{\star}$ unchanged—so all convergence results are stated for $X^{\star}$ rather than for the factors or the loadings. This identification issue is revisited in Section 3.4.

3.2 Assumptions

Assumption 1 (Boundedness and noise)**.** *The factors $\{F^{\star}_t\}_{t=1}^T$ are supported on the compact box $\mathcal{B} = [-B, B]^K$ with $B \geq 1$ almost surely. Conditional on $F^{\star}$, $\mathrm{vec}(U) \overset{d}{=} \Sigma^{1/2}\mathrm{vec}(Z)$, where $Z \in \mathbb{R}^{N \times T}$ has independent, zero-mean, sub-Gaussian entries with sub-Gaussian norm bounded by σ_z, and Σ is positive semidefinite with bounded spectral norm.*

Assumption 2 (Smoothness)**.** *There exists $\beta \in \mathbb{N}_+$ with $\beta \geq 1$ such that $\varphi^{\star}_i \in \mathcal{H}^{\beta}(\mathcal{B} \times [0, 1], B)$ for each i.*

Assumption 3 (Encoder approximation class)**.** *There exist a sparse linear map* $W^\star \in \mathbb{R}^{K\times N}$ *with* $\|W^\star\|_0 \le L_N K$ *and* $\|W^\star\varphi^\star\|_\infty \le B$*, and a map* $\eta^\star : [-2B, 2B]^K \times [0,1] \to \mathbb{R}^K$ *with* $\eta^\star \in (\mathcal{H}^\beta([-2B,2B]^K \times [0,1], C_\eta))^K$ *for a finite constant* $C_\eta = C_\eta(\beta, K, B)$ *that is* uniform in N*, such that*

$$\sup_{(x,s)\in\mathcal{B}\times[0,1]} \|\eta^\star(W^\star\varphi^\star(x,s),s) - x\|^2 + \|W^\star\|^2 \;\le\; 2\epsilon_N. \tag{3}$$

Assumption 4 (Weak dependence)**.** *The process* $\{(F_t^\star, U_t)\}_{t=1}^\infty$ *is strictly stationary,* β*-mixing with* $\beta(u) \le e^{-\kappa u}$ *for some* $\kappa > 0$*, and* ρ*-mixing with* $\sum_{u=1}^\infty \rho(u) < \infty$*.*

3.3 Autoencoder architectures

3.3.1 The disjoint-output autoencoder

Let $\sigma(x) = \max(0, x)$ denote the ReLU activation. Following Shen and Xiu (2026), for depth $d \in \mathbb{N}$, width $w \in \mathbb{N}$, and dimension vector $n = (n_0, \ldots, n_{d+1})$, let $\mathcal{F}_{n_0}^{n_{d+1}}(d, w, C, B)$ denote the class of fully connected feedforward networks $f : \mathbb{R}^{n_0} \to \mathbb{R}^{n_{d+1}}$ of the form

$$f(x) = W^{(d)}\sigma_{v^{(d)}}W^{(d-1)}\sigma_{v^{(d-1)}}\cdots W^{(1)}\sigma_{v^{(1)}}W^{(0)}x, \tag{4}$$

where each $W^{(\ell)}$ is an $n_{\ell+1} \times n_\ell$ weight matrix, $v^{(\ell)}$ is a shift vector, and the parameters satisfy $\max_\ell \|W^{(\ell)}\|_\infty \vee \|v^{(\ell)}\|_\infty \le C$ with $\|f\|_\infty \le B$.

3.3.2 The global time-conditioned autoencoder

The primary estimator of this paper is the *global time-conditioned autoencoder* (TV-AE-G). It is a single autoencoder in which the scaled time index $s = t/T$ enters as an additional input to both the encoder and the decoder. The encoder and decoder classes are

$$\rho \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, C, B), \qquad \psi_i \in \mathcal{F}_{K_1+1}^{1}(d_2, w_2, C, B), \tag{5}$$

and the model is

$$X_{it} \;=\; \psi_i(\rho(X_t, t/T),\, t/T) + U_{it}. \tag{6}$$

The estimator minimizes the unweighted loss

$$(\hat\rho, \{\hat\psi_i\}_{i=1}^N) = \arg\min \sum_{t=1}^T \sum_{i=1}^N \left(X_{it} - \psi_i(\rho(X_t, t/T), t/T)\right)^2, \tag{7}$$

over $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B)$, where $\mathcal{F}_{AE}^{K_1}$ is the disjoint-output class of Shen and Xiu (2026) extended to $(N+1)$- and (K_1+1)-dimensional inputs. The estimated common component is $\hat X_{it}^G := \hat\psi_i(\hat\rho(X_t, t/T), t/T)$.

The phrase "shares parameters across time" is used in a precise sense. It does *not* mean that the encoder and decoder are time-invariant: their outputs depend on s, and so they vary with t. It means that the *network weights* are common to all t, and that the time variation is carried by the input s rather than by a time-specific set of weights. Formally, there is a single map $\rho(\cdot,\cdot)$ and a single map $\psi_i(\cdot,\cdot)$ per unit, evaluated at $(X_t, t/T)$ and $(\rho(X_t, t/T), t/T)$

respectively. An analogy from linear models: a varying-coefficient regression models the coefficient as $\beta_t = \beta(t/T)$ for a single smooth function $\beta(\cdot)$; the parameters of $\beta(\cdot)$ are shared across t, but the value $\beta(t/T)$ differs at each t. The map $\rho(\cdot,\cdot)$ plays the same role as $\beta(\cdot)$.

The contrast with the local autoencoder of Section D is sharp. In TV-AE-L, each of m grid points has its own $(\rho_j, \{\psi_{i,j}\})$; the parameter count is of order $m(|\rho| + N|\psi|)$ and the effective sample per fit is only Th. In TV-AE-G, there is one $(\rho, \{\psi_i\})$ with parameter count $|\rho| + N|\psi|$ and effective sample T. This is what makes TV-AE-G statistically tractable at finite T, and, as Theorem 1 shows, it is also what makes it rate-optimal: the global estimator attains the full $(K+1)$-dimensional nonparametric rate, while the local estimator pays a kernel-smoothing penalty.

3.3.3 The characteristic-conditioned autoencoder

For applications to portfolio management and asset pricing, we generalize TV-AE-G to condition on observable characteristics in the spirit of Gu, Kelly, and Xiu (2021, GKX). Let $Z_{i,t-1} \in \mathbb{R}^P$ denote unit i's characteristics measured at $t-1$. The *characteristic-conditioned time-varying autoencoder* (TV-AE-C) is

$$r_{it} \;=\; \psi_i(\rho(X_t, t/T),\, Z_{i,t-1},\, t/T) + u_{it}, \tag{8}$$

where $\rho : \mathbb{R}^N \times [0,1] \to \mathbb{R}^{K_1}$ is the encoder and each $\psi_i : \mathbb{R}^{K_1} \times \mathbb{R}^P \times [0,1] \to \mathbb{R}$ is a decoder. The target loading function is $\varphi_i^\star : \mathcal{B} \times [-B_Z, B_Z]^P \times [0,1] \to \mathbb{R}$, assumed β-Hölder in its arguments, with $\|Z_{i,t-1}\|_\infty \le B_Z$ almost surely.

Model (8) nests GKX as the special case with no t/T input, TV-AE-G as the special case with $P = 0$, and a static linear factor model when both inputs are dropped and the decoders are linear.

3.4 Identification

The factors and loading functions in (2) are not separately identified. For any time-indexed family of invertible transformations $\mu_t : \mathbb{R}^K \to \mathbb{R}^K$, the reparameterization

$$F_t \mapsto \mu_t(F_t), \qquad \varphi_i^\star(\cdot, t/T) \mapsto \varphi_i^\star(\mu_t^{-1}(\cdot), t/T) \tag{9}$$

leaves the common component $X_{it}^\star$ unchanged. This generalizes the rotation indeterminacy of static linear factor models: μ_t may now be a nonlinear and time-varying map. Two consequences are important for the analysis that follows.

1. The *common component* $X_{it}^\star$ is identified, and all our convergence results are stated directly for $X_{it}^\star$.

2. The *factor series* $\{\hat{F}_t\}$ is identified only up to a time-varying invertible transformation. Comparisons of $\hat{F}_t$ to $F_t^\star$ therefore require a nonlinear alignment procedure or a rotation-invariant metric. We adopt the common-component mean-squared error as the primary metric throughout, precisely because it sidesteps this ambiguity.

4 Theory

This section states the main theoretical results for TV-AE-G and TV-AE-C. All proofs are collected in the appendices. Throughout, we write $a_{N,T} \lesssim b_{N,T}$ when $a_{N,T} \le C b_{N,T}$ for a constant C independent of N and T, and we set $s_t = t/T$ and $X^\star_{it} = \varphi^\star_i(F^\star_t, s_t)$.

4.1 In-sample rate for TV-AE-G

Theorem 1 (In-sample recovery, TV-AE-G)**.** *Suppose Assumptions 1–3 hold and* $\log T = o(\epsilon_N^{-1})$*. Let* $K \le K_1 \le \min(w_1, w_2)$*,* $d_1 \asymp d_2 \asymp \log T$*, and* $w_1 \asymp w_2 \asymp T^{(K+1)/(2(2\beta+K+1))}$*. Then for sufficiently large* $\min(N,T)$*, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{NT}\sum_{t=1}^{T}\sum_{i=1}^{N}\left(\hat{X}^{\mathrm{G}}_{it} - X^\star_{it}\right)^2 \;\lesssim\; \left(T^{-\frac{2\beta}{2\beta+K+1}} \;+\; N^{-1}K_1 \;+\; \epsilon_N\right)\log^3 T. \tag{10}$$

The rate decomposes into three terms. The first, $T^{-2\beta/(2\beta+K+1)}$, is the nonparametric rate for estimating the loading function $\varphi^\star_i(F,s)$ in $K+1$ effective dimensions. The second, $N^{-1}K_1$, is the parametric estimation error for the TK_1 unknown factor values. The third, ϵ_N, is the encoder approximation error from Assumption 3.

Remark 1 (Comparison with the local estimator)**.** *The direct nonlinear analogue of L. Su and Wang (2017) replaces the global loss* (7) *with a kernel-weighted loss at each of* $m \asymp 1/h$ *grid points. The resulting local autoencoder (*TV-AE-L*) attains the rate* $(Th)^{-2\beta/(2\beta+K+1)} + N^{-1}K_1 + \epsilon_N$ *at each grid point, and optimizing over the bandwidth yields the best achievable rate* $T^{-2\beta/(2\beta+K+2)}$ *(Appendix D). The gap between* $T^{-2\beta/(2\beta+K+1)}$ *and* $T^{-2\beta/(2\beta+K+2)}$ *is the kernel-smoothing penalty of local estimation, and it is the reason the empirical work of Section 6 uses* TV-AE-G *rather than* TV-AE-L*. As* $T \to \infty$*, the two rates converge to zero at different polynomial orders; for any fixed* $T > 1$ *the gap is* $T^{-(2\beta)/((2\beta+K+1)(2\beta+K+2))}$*, which grows with* K *and* β*.*

Remark 2 (Sharpness of the log factor)**.** *The* $\log^3 T$ *factor in* (10) *is not the* $\log^4 T$ *of Shen and Xiu (2026). The improvement comes from a sharper tracking of the entropy term* $T^{(K+1)/(2\beta+K+1)}$ *in the empirical-process concentration step (Lemma 9 below) and of the factor net* $\mathcal{S}$*. The polynomial rate is unaffected either way; we record the sharper power for completeness.*

4.2 The characteristic-conditioned estimator

Proposition 1 (In-sample recovery, TV-AE-C)**.** *Suppose the conditions of Theorem 1 hold. Suppose in addition that* $\|Z_{i,t-1}\|_\infty \le B_Z$ *almost surely and that the target loading function* $\varphi^\star_i : \mathcal{B} \times [-B_Z, B_Z]^P \times [0,1] \to \mathbb{R}$ *lies in* $\mathcal{H}^\beta(\mathcal{B} \times [-B_Z, B_Z]^P \times [0,1], B)$ *for each* i*. Let the decoder class be* $\mathcal{F}^1_{K_1+P+1}(d_2, w_2, C, B)$*, with* $d_2 \asymp \log T$ *and* $w_2 \asymp T^{(K+P+1)/(2(2\beta+K+P+1))}$*. Then for sufficiently large* $\min(N,T)$*, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{NT}\sum_{t,i}\left(\hat{X}^{\mathrm{C}}_{it} - X^\star_{it}\right)^2 \;\lesssim\; \left(T^{-\frac{2\beta}{2\beta+K+P+1}} \;+\; N^{-1}K_1 \;+\; \epsilon_N\right)\log^3 T, \tag{11}$$

where $\hat{X}_{it}^{C} := \hat{\psi}_i(\hat{\rho}(X_t, s_t), Z_{i,t-1}, s_t)$ *is the* TV-AE-C *reconstruction.*

Remark 3 (Cost of conditioning)**.** *Proposition 1 recovers Theorem 1 when* $P = 0$*. The* P*-dimensional cost is* $T^{-2\beta/(2\beta+K+P+1)}$ *instead of* $T^{-2\beta/(2\beta+K+1)}$*. For the empirical work of Section 6 (*$P = 6$*),* $\beta = 2$*,* $K = 2$*, this is* $T^{-0.36}$ *vs.* $T^{-0.57}$*: a real slowdown. The tradeoff is that the characteristics may absorb part of the loading-function variation that would otherwise be attributed to the nonlinear factor structure. If some components of* $Z_{i,t-1}$ *are discrete or lie on a lower-dimensional subset, the effective dimension is smaller than* P *and the rate is correspondingly faster. The choice between* TV-AE-G *and* TV-AE-C *is thus a bias-variance tradeoff in the conditioning dimension, not a strict improvement.*

4.3 Factor recovery

Proposition 2 (Factor recovery for TV-AE-G)**.** *Under the conditions of Theorem 1, there exists, for each* $t = 1, \ldots, T$*, a map* $\mu_t : \mathbb{R}^{K_1} \to \mathbb{R}^K$ *(depending on the trained encoder, the architecture, and the DGP, but not on data realizations beyond the encoder) such that, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{T}\sum_{t=1}^{T} \|\mu_t(\hat{\rho}(X_t, s_t)) - F_t^\star\|^2 \;\lesssim\; \left(\epsilon_N K_1 \;+\; N\epsilon_N T^{-\frac{2\beta}{2\beta+K+1}} \;+\; N\epsilon_N^2\right)\log^4 T. \tag{12}$$

Remark 4. *The map* μ_t *is not the identity in general. The estimated factors* $\hat{\rho}(X_t, s_t)$ *are identified only up to a nonlinear, time-indexed transformation. This is the same indeterminacy as in Shen and Xiu (2026)'s Theorem 3, extended to the time-varying case. It is a genuine feature of the nonlinear model and not an artifact of the estimator. In the linear case with* $\varphi_i^\star(F, s) = \lambda_i(s)^\top F$ *and* $L_N \asymp N$*,* $\epsilon_N \lesssim N^{-1}$ *and the rate simplifies to* $N^{-1}K_1 + T^{-2\beta/(2\beta+K+1)}$*, matching Shen and Xiu (2026) up to the extra time dimension.*

4.4 Discussion

The theory has three implications for practice.

First, the global estimator dominates the local estimator in rate. The gap $T^{-2\beta/(2\beta+K+1)}$ vs. $T^{-2\beta/(2\beta+K+2)}$ is the kernel-smoothing penalty, and it exists because the local estimator must estimate the loading function at each reference time from only Th observations. The global estimator pools information across all T observations and pays no such penalty. This is the technical justification for the empirical design in Section 6.

Second, the rate depends on the factor dimension K and the nonlinearity smoothness β in a specific way. Larger K increases the nonparametric difficulty: the loading function lives on a $(K+1)$-dimensional domain, and the exponent $2\beta/(2\beta+K+1)$ decreases accordingly. Higher smoothness β increases the exponent and accelerates convergence. This is the classical nonparametric tradeoff, and it is why the autoencoder is better suited to panels with low-dimensional nonlinear structure (crypto returns) than to panels with high-dimensional or nearly-linear structure (equity returns).

Third, the encoder approximation error ϵ_N does not depend on the network architecture. This is the sense in which TV-AE-G is "architecture-free" asymptotically: the number

of encoder parameters does not affect the convergence rate, provided the encoder class is rich enough to approximate $\eta^\star$ in Assumption 3. In practice, architecture affects the finite-sample performance, and this is what the simulations in Section 5 and the empirical work in Section 6 document.

5 Simulation study

We evaluate TV-AE-G against a set of benchmarks on a family of simulated panels whose data-generating processes span the two axes of complexity that motivate the paper: nonlinearity in the loading function, and time variation in its argument.

5.1 Design

We simulate panels $X \in \mathbb{R}^{N\times T}$ from the factor model (2). The factor dimension is $K = 2$, the cross-sectional size is $N = 100$, and the time-series length is $T \in \{200, 500, 1000, 2000\}$. The latent factors $F_t^\star \in \mathbb{R}^K$ are drawn independently across t from the uniform distribution on $[-2, 2]^K$.

Full design and reported subset. The full factorial design crosses five loading-function classes (linear, exponential, quadratic, radial, sigmoid) with five time-variation schemes (constant, single abrupt break, multiple abrupt breaks, smooth logistic, smooth double logistic) and three noise structures (iid, heteroskedastic, cross-sectionally dependent), giving $5 \times 5 \times 3 = 75$ DGPs. The main text reports five DGPs that isolate the nonlinearity and time-variation channels: linear/constant, exponential/constant, linear/logistic, exponential/logistic, and quadratic/logistic. The heteroskedastic and cross-sectionally dependent noise variants were run on a subset and produced qualitatively identical conclusions; they are omitted from the main tables because they do not change the comparison. The full grid is reported in a companion note.

Time variation. The time-varying case uses $\lambda_i(t/T) = \lambda_{i0} + b\, g(t/T)$, where g is the logistic transition $g(s) = 1/(1 + \exp(-\kappa(s - 1/2)))$ with $\kappa = 10$, and $b = 1$. The constant case ($g \equiv 0$) plays the role of a placebo test.

Replication counts. Each DGP is simulated 10 times at $T = 200$, 5 times at $T = 500$ and $T = 1000$, and 3 times at $T = 2000$ (the quadratic DGP only). The replication counts are chosen to balance Monte Carlo error against computation; the autoencoder training is the dominant cost at $T = 1000$ and $T = 2000$.

Each unit's common component is normalized to unit variance, and the idiosyncratic noise U_{it} is i.i.d. standard normal in the main-text runs.

5.2 Methods and implementation

We compare four estimators.

Static PCA. The static linear factor model $X_{it} = \lambda_i' F_t + U_{it}$, estimated by PCA on the full training sample with $K_1 = 5$ factors. This is the standard benchmark in the factor-model literature.

Local PCA (LPCA). The L. Su and Wang (2017) estimator with an Epanechnikov kernel and Silverman's rule-of-thumb bandwidth. At each grid point, a kernel-weighted eigenvalue decomposition is computed; the reconstruction uses the second-stage refit of L. Su and Wang (2017). This is the benchmark against which the time-varying autoencoder is primarily compared.

Static autoencoder (StaticAE). The disjoint-output autoencoder of Shen and Xiu (2026) with no time input. The model is $X_{it} = \psi_i(\rho(X_t)) + U_{it}$, with $\rho : \mathbb{R}^N \to \mathbb{R}^{K_1}$ and each $\psi_i : \mathbb{R}^{K_1} \to \mathbb{R}$. This isolates the contribution of the nonlinear encoder from the contribution of the time input.

Global time-conditioned autoencoder (TVAEG). A single autoencoder with both encoder and decoder conditioned on t/T, $X_{it} = \psi_i(\rho(X_t, t/T), t/T) + U_{it}$.

Local time-varying autoencoder (TVAEL). A direct nonlinear analogue of local PCA, described in Appendix D. At each grid point, a separate autoencoder is trained on the kernel-weighted panel; reconstruction at t averages across grid points with the same kernel weights. This estimator is included in a separate set of runs (Section 5.3) but is not part of the main comparison table because it did not train at the sample sizes we consider.

Hyperparameters. All four autoencoder estimators use the disjoint-output architecture of Shen and Xiu (2026) and are trained by stochastic gradient descent with the Adam optimizer. The final configuration uses bottleneck dimension $K_1 = 3$, encoder width 64, decoder width 32, $n_{\text{epochs}} = 500$, early-stopping patience 30, learning rate 10^{-3}, weight decay 10^{-3}, and ensemble size $n_{\text{seeds}} = 3$. These values were selected by the capacity sweep reported in Section 5.6. Both neural estimators use the same architecture, initialization, optimizer, weight decay, batch size, early-stopping rule, and ensemble size. The only difference between them is the time input. This isolation is essential for attributing the observed performance differences to the correct source.

5.3 The local autoencoder at finite T

A local autoencoder analogue of L. Su and Wang (2017) was implemented and evaluated on the same grid. At $T = 200$ and $N = 100$, with the Silverman bandwidth $h \approx 0.148$, the effective sample at each grid point is $Th \approx 30$ observations. A network with roughly 8,000 parameters trained on 30 observations does not learn. In the first TVAEL run (10 grid points, 2 seeds, 300 epochs), the estimator produced an MSE of 0.1000 on the linear/constant DGP (vs. 0.0483 for StaticAE) and 0.1885 on the quadratic/logistic DGP (vs. 0.1701 for StaticAE). That is, TVAEL underperformed both the static autoencoder and, on some DGPs, the linear methods.

Table 1: Mean squared error of the common component across simulation replications. Lower is better.

DGP	T	PCA	LPCA	StaticAE	TVAEG
Linear, constant	200	0.0344	0.0671	0.0482	0.0476
Linear, constant	500	0.0256	0.0422	0.0324	0.0326
Linear, constant	1000	0.0235	0.0333	0.0276	0.0277
Exponential, logistic	200	0.1919	0.1775	0.1116	0.0875
Exponential, logistic	500	0.1830	0.1536	0.0722	0.0589
Exponential, logistic	1000	0.2018	0.1598	0.0500	0.0458
Quadratic, logistic	200	0.4604	0.4415	0.1601	0.1460
Quadratic, logistic	500	0.4306	0.4394	0.1054	0.0928
Quadratic, logistic	1000	0.4475	0.4614	0.0860	0.0870
Quadratic, logistic	2000	0.4586	0.4424	0.0699	0.0791

Diagnostics confirmed the cause. Effective training sample per grid was 48 observations in the interior and 7 at the boundaries. Early stopping fired at 60–136 epochs. The model was not overtrained; it was undertrained due to data scarcity. This is a fundamental property of local nonparametric estimation at $T = 200$, not a tuning issue, and it is consistent with the rate comparison in Remark 1: the local estimator pays a kernel-smoothing penalty, and at $T = 200$ the penalty dominates.

This finding is the practical motivation for the global estimator. The rate result (Theorem 1) says that the global estimator is asymptotically faster; the simulation evidence says that at finite T the gap is large enough that the local estimator is essentially unusable at the sample sizes of interest.

5.4 Results

Table 1 reports the mean MSE across replications for the four main estimators.

Three findings stand out. First, on linear, constant, where the true loading function does not depend on t, TV-AE-G and StaticAE are indistinguishable at every sample size. This is the correct behavior: when there is no time variation, the extra input carries no information, and the estimator cannot benefit from it. Second, on exponential, logistic, TV-AE-G beats StaticAE by 21.6% at $T = 200$, 18.4% at $T = 500$, and 8.4% at $T = 1000$; against PCA, the gap is an order of magnitude larger, and PCA plateaus at ≈ 0.19–0.20 across all sample sizes, reflecting its irreducible misspecification bias. Third, on quadratic, logistic, the direction reverses as T grows: TV-AE-G beats StaticAE at $T \leq 500$ and is overtaken at $T = 2000$. This is the rate-variance tradeoff predicted by the theory: the dominant nonlinearity is time-invariant, and the time input's bias correction is eventually outweighed by its variance cost.

5.5 The symmetric-regularization correction

An earlier specification inadvertently applied weight decay to TV-AE-G but not to StaticAE, producing a spurious 25% advantage for TV-AE-G on the linear/constant placebo DGP

at $T = 500$ and $T = 1000$. Since the DGP has no time variation, the time input cannot genuinely help, so the advantage had to be an artifact. The cause was located in the training code: the weight-decay parameter was passed to the TV-AE-G optimizer but not to the StaticAE optimizer. Under the corrected symmetric specification, the two estimators tie on the placebo DGP (Table 1), which is the correct result. We report the correction because it illustrates two things: the placebo test is informative and can detect implementation asymmetries that a comparison on the target DGP would miss; and comparisons between regularized and unregularized models are meaningless, so any paper claiming an autoencoder advantage must demonstrate that the benchmark is regularized comparably.

5.6 Capacity sweep

The scaling study revealed that early stopping was firing at $T = 200$ because the validation loss was noise-dominated. A capacity sweep was run on the exponential/logistic and quadratic/logistic DGPs (the two DGPs where the autoencoder has the largest theoretical advantage) to select the final hyperparameters. The sweep varied weight decay $\lambda \in \{0, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$, decoder width $\in \{16, 32, 64\}$, encoder width $\in \{32, 64, 128\}$, and bottleneck dimension $K_1 \in \{2, 3, 4\}$. The findings:

- weight decay: monotone improvement up to 10^{-3}, monotone degradation above 5×10^{-3}. The optimal value is 10^{-3};
- decoder width: flat, with slightly better performance at widths 32 and 64 than at 16;
- encoder width: flat, with slightly better performance at widths 64 and 128 than at 32;
- bottleneck: $K_1 = 3$ dominates $K_1 = 2$ consistently; $K_1 = 4$ is comparable but not better.

The final configuration ($K_1 = 3$, encoder 64, decoder 32, weight decay 10^{-3}) is the one reported in Table 1.

5.7 What the simulations establish

The simulations establish three findings. First, the autoencoder recovers the common component at the rate predicted by Theorem 1. On the placebo DGP, it matches the linear estimator within noise; on the exponential/logistic DGP, it dominates both linear and static-autoencoder benchmarks, and the proportional advantage is stable across T. Second, the autoencoder's advantage over the correctly specified linear estimator grows with the amount of nonlinearity in the loading function. Third, the advantage is not unlimited: on the quadratic/logistic DGP, where the dominant nonlinearity is time-invariant, TV-AE-G is eventually overtaken by StaticAE, and this is the correct behavior of the estimator under the bias-variance tradeoff.

6 Empirical analysis

6.1 Design and rationale

The empirical work in this paper evaluates the theory on real panels. The theory (Theorem 1) constrains the accuracy with which TV-AE-G recovers the common component $X^\star$. The natural empirical test is therefore a comparison of out-of-sample reconstruction accuracy across panels with varying degrees of nonlinearity in the loading structure.

The empirical section is organized around three applications, in order of how directly each tests the theory.

The *reconstruction* R^2 comparison (Section 6.2) is the direct test. It measures $\hat{X}_{it}$ against X_{it} on held-out data, which is the quantity the theory bounds. It is also the quantity that Shen and Xiu (2026) and L. Su and Wang (2017) report, so the comparison against those benchmarks is apples-to-apples.

The *minimum-variance portfolio* application (Section 6.3) is a downstream application the theory does not address. MVP Sharpe depends on the small eigenvalues of the inverse covariance matrix, which are not the quantities the theory promises to recover. It is included because it is a standard benchmark in the covariance-estimation literature, not because the theory predicts the outcome.

The *equity return prediction* application was attempted and did not produce a competitive result at the sample sizes we have access to. It is documented in Section 6.4 along with the diagnostics that explain why, because the failure is informative about what the method can and cannot do.

A fourth piece, the *local autoencoder and local PCA* at the rolling-window sample sizes, is also documented in Section 6.4: the effective sample per grid point is too small for these methods to train at $T = 250$.

6.2 The reconstruction application

6.2.1 Data

We use four panels.

Equity returns. Daily returns for the 100 largest S&P 500 constituents by average market capitalization over 2005–2024, from Refinitiv Datastream. The panel has 5,029 trading days and 100 stocks, with coverage averaging 0.62 across the (i, t) grid (firms enter and exit the universe as index composition changes, and a small number of stocks have missing observations).

Equity volatility. For the same 100 firms, daily log realized volatility $\log \mathrm{RV}_{it} = \frac{1}{2} \log(\sum_{s=0}^{4} r_{i,t-s}^2 + \epsilon)$, with $\epsilon = 10^{-10}$ protecting against log of zero. This panel has substantially higher signal-to-noise than the return panel and is the canonical setting for the linear factor structure documented in the stochastic-volatility literature.

Crypto returns. Daily returns for 50 cryptocurrencies, 2015–2024, from Refinitiv. The universe was constructed as follows. The composite crypto feed ({COIN}=) is available for five majors (BTC, ETH, XRP, LTC, BCH). The remaining coins required exchange-specific suffixes. Three suffix conventions were tested: {COIN}= (composite), {COIN}=CCCL (CoinDesk composite), and {COIN}=BTSP (Bitstamp). Fifty of 58 candidate coins resolved under at least one convention; the remaining 8 (EOS, KSM, ZIL, WAVES, ANKR, STORJ, CELO, EURT) did not resolve under any tested suffix and were dropped. Coverage builds over time as new coins are listed; the panel reaches 25 coins by mid-2021 and 50 coins by December 2021. Because of this coverage profile, the out-of-sample window is June 2021 to December 2024.

Crypto volatility. For the same 50 coins, daily log realized volatility, constructed as for the equity volatility panel.

6.2.2 Estimation

We evaluate rolling out-of-sample R^2 with an estimation window of $T_{\text{est}} = 250$ trading days and refits every $H = 21$ trading days. The pooled out-of-sample R^2 is

$$R^2 = 1 - \frac{\sum_{(i,t)\in\text{OOS}}(X_{it} - \hat{X}_{it})^2}{\sum_{(i,t)\in\text{OOS}} X_{it}^2}. \tag{13}$$

The refit is repeated through the out-of-sample period; for the equity panels the out-of-sample period is 2010–2024 (70 refit windows); for the crypto panels it is 2021-06 to 2024-12 (roughly 70 refit windows, depending on the exact first refit date).

The unbalanced panel. The equity and crypto panels are both unbalanced. To accommodate a variable number of units at each date, each permno (or coin) is assigned a fixed slot in the encoder input of dimension $N_{\max} = 100$ (equity) or $N_{\max} = 50$ (crypto). Missing values are filled with zero, and a 0/1 mask vector is concatenated with the returns and the time index, giving an encoder input of dimension $2N_{\max} + 1$.

The PCA warm-start. The encoder is parameterized as

$$F_t = W_{\text{PCA}} X_t + \text{MLP}(X_t), \tag{14}$$

where $W_{\text{PCA}} \in \mathbb{R}^{K_1 \times N}$ is initialized to the top-K_1 PCA loadings of the training window and the MLP output layer is initialized to zero. At initialization, the encoder exactly outputs PCA; training can either preserve this initialization (if the loss does not decrease) or improve on it. The weight decay on W_{PCA} is set to zero, so the PCA backbone is not shrunk. This warm-start is necessary in the empirical applications (though not in the simulations, where signal-to-noise is higher): without it, gradient descent on monthly or daily returns collapses to the per-stock mean, because the signal-to-noise ratio at those frequencies is roughly 10^{-4}. The collapse and its diagnosis are documented in Section 6.4.

The autoencoder hyperparameters are the same as in the simulations ($K_1 = 5$ for the reconstruction application, encoder width 64, decoder width 32, learning rate 10^{-3}, weight

Table 2: Out-of-sample reconstruction R^2 on four panels. $N = 100$ stocks or $N = 50$ coins; $T_{\text{est}} = 250$; $H = 21$.

Panel	PCA	StaticAE	TVAEG
Equity returns	0.578	0.574	0.573
Equity volatility	0.989	0.989	0.985
Crypto returns	0.606	0.626	0.642
Crypto volatility	0.972	0.972	0.966

Table 3: Paired bootstrap over out-of-sample refit windows on the cryptocurrency return panel (2,000 draws).

Comparison	ΔR^2	95% CI
TV-AE-G − PCA	+0.036	[+0.008, +0.075]
TV-AE-G − StaticAE	+0.015	[−0.001, +0.045]
StaticAE − PCA	+0.021	[+0.007, +0.036]

decay 10^{-3}, 200 epochs, ensemble of 3 seeds), except that $K_1 = 5$ rather than $K_1 = 3$ because the reconstruction application has more cross-sectional variation.

6.2.3 Results

Table 2 reports the pooled out-of-sample R^2 for each method on each panel.

Three features stand out.

On the equity panels, the autoencoder is statistically indistinguishable from PCA (the differences are less than one percentage point, well within Monte Carlo noise). The linear factor structure is not misspecified on these panels, so the autoencoder's extra flexibility is harmless but not useful.

On the crypto volatility panel, the same holds: both methods achieve R^2 above 0.97 and the difference is not statistically distinguishable from zero. Crypto volatility, despite being from the same coins as crypto returns, is dominated by a small number of linear factors.

On the crypto return panel, the ordering changes. StaticAE gains 2.0 percentage points over PCA, and TV-AE-G gains another 1.6 points over StaticAE, for a total of 3.6 points over PCA. This is the panel where the loading function is genuinely nonlinear, and it is the panel where the autoencoder wins.

6.2.4 Bootstrap inference

To assess whether the crypto-return improvement is statistically significant, we perform a paired bootstrap over the 70 refit windows. The bootstrap resamples the windows with replacement 2,000 times, recomputes the pooled R^2 for each method on the resampled set, and reports the distribution of the difference.

Table 3 reports the results. Two of the three comparisons exclude zero at the 5% level: the autoencoder beats PCA by 2–4 points, and TV-AE-G beats PCA by 3.6 points. The TV-AE-G vs. StaticAE comparison has a point estimate of +0.015 and a CI that includes

Table 4: Out-of-sample reconstruction R^2 by year, crypto return panel.

Year	PCA	StaticAE	TV-AE-G
2021	0.666	0.712	0.784
2022	0.630	0.639	0.638
2023	0.509	0.544	0.544
2024	0.604	0.608	0.607

Table 5: Distribution of out-of-sample reconstruction R^2 across the 70 refit windows, crypto return panel.

Method	Mean	Median	Std. dev.	Min	Max
PCA	0.601	0.613	0.178	0.126	0.976
StaticAE	0.623	0.625	0.157	0.143	0.973
TV-AE-G	0.626	0.630	0.158	0.144	0.974

zero marginally, so the evidence for the time input over the static autoencoder is directional but not decisive at this sample size.

6.2.5 By-year breakdown

The pooled comparison masks year-to-year variation. Table 4 disaggregates the crypto-return results by year.

The autoencoder beats PCA in every year. The margin is small in 2022–2024 (0.4 to 3.5 percentage points) and large in 2021 (7.2 to 11.8 points). The 2021 concentration is consistent with the theory: the time input matters most when the loading structure is shifting, and 2021 was the most pronounced crypto regime shift in the sample.

6.2.6 Per-window distribution

The pooled R^2 could in principle be driven by a small number of outlier windows. Table 5 reports the distribution of R^2 across the 70 refit windows.

The autoencoder methods have both higher mean and higher median than PCA, so the pooled improvement is not driven by outliers. They also have lower standard deviation across windows (0.157–0.158 vs. 0.178), so they are more stable. This is a stronger statement than the pooled comparison alone and is a useful complement to the bootstrap.

6.2.7 Bottleneck-width sensitivity

The autoencoder's advantage over PCA peaks at $K_1 = 2$ (a gain of 3.4 percentage points over PCA) and declines smoothly as K_1 grows, collapsing to zero at $K_1 = 12$. This pattern is consistent with the interpretation that crypto returns have genuine nonlinear structure in a moderate number of dimensions, and that PCA's extra factors eventually absorb it once K_1 is large enough. The sensitivity is also consistent with the theory's prediction that K_1 should not be over-specified: the $N^{-1}K_1$ term in the rate is the parametric cost of estimating more factor values than necessary.

6.2.8 The one-sided test for nonlinearity

The comparison across panels can be framed as a one-sided test for the presence of nonlinearity in the loading function. Under the null that the loading function is linear, the autoencoder is over-parameterized, so its expected out-of-sample R^2 is weakly lower than PCA's. Under the alternative, the autoencoder can capture the nonlinear structure, so its expected R^2 can be higher. The test is:

- reject linearity if the bootstrap CI on $\Delta R^2 = R^2_{\text{AE}} - R^2_{\text{PCA}}$ is entirely above zero;
- fail to reject otherwise.

The test is asymmetric: rejection is evidence of nonlinearity, non-rejection is not evidence of linearity. Non-rejection can arise because the loading function is linear, because the autoencoder is underpowered at the given sample size, or because the nonlinearity is too small to detect.

On the four panels, we obtain non-rejection on equity returns, equity volatility, and crypto volatility, and rejection on crypto returns ($\Delta R^2 = +0.036$, 95% CI $[+0.008, +0.075]$). The crypto return panel is the only one where we can reject linearity at the 5% level.

The test has a known limitation: the alternative includes both time-varying linear loadings and nonlinear loadings, and at this sample size we cannot distinguish between the two channels. If the true loading function is $\lambda_i(t/T)'F_t$ (time-varying but linear) and the time variation is strong enough, TV-AE-G can reject linearity by capturing the time variation rather than the nonlinearity. The local PCA analogue would distinguish the channels, but the local estimators do not train at $T = 250$ (Section 6.4).

6.3 Minimum-variance portfolios

6.3.1 Setup

The MVP weights depend only on the covariance matrix:

$$w = \frac{\hat{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}'\hat{\Sigma}^{-1}\mathbf{1}}, \tag{15}$$

and the MVP Sharpe ratio depends on how well $\hat{\Sigma}$ approximates the true covariance. We use the S&P 100 daily panel (497 stocks, 5,029 trading days, 2005–2024), with cleaning that drops rows with missing return or market cap, converts Refinitiv percent returns to decimal, and drops rows with $|r_{it}| \geq 50\%$ (almost certainly data errors in daily equity data). Estimation is by a rolling window of $T = 250$ days with monthly rebalancing ($H = 21$ days). The out-of-sample period is 2010–2024.

We compare nine estimators: equal-weight, sample covariance, Ledoit–Wolf linear shrinkage, Ledoit–Wolf nonlinear shrinkage (OAS), POET, FWYZ (the time-varying local PCA covariance estimator of **fan2024time**), the static autoencoder with POET residual covariance, TV-AE-G with POET residual covariance, and TV-AE-C with $K_1 \in \{3, 5\}$.

Table 6: Out-of-sample Sharpe ratios for minimum-variance portfolios on the S&P 100 panel. $T_{\text{est}} = 250$, $H = 21$, 2010–2024.

Estimator	Sharpe	Ann. vol.
Equal weight (1/N)	0.838	0.173
Sample covariance	0.855	0.142
Ledoit–Wolf linear	**1.005**	0.128
Ledoit–Wolf nonlinear (OAS)	0.977	0.131
POET	0.984	**0.124**
StaticAE (warm) + POET	0.761	0.121
TV-AE-G (warm) + POET	0.733	0.121
TV-AE-C ($K_1 = 5$) + POET	0.733	0.132
TV-AE-C ($K_1 = 3$) + POET	0.708	0.131
S&P 500	0.730	0.172

6.3.2 Results

Table 6 reports annualized Sharpe ratios over the out-of-sample period.

The best performer is Ledoit–Wolf linear shrinkage (1.005), followed by POET (0.984). The autoencoder-based covariances deliver Sharpe ratios of 0.71–0.76, below the linear shrinkage benchmark and roughly comparable to the equal-weighted portfolio. The autoencoder recovers the factor structure (the volatility is in line with PCA and POET, confirming that the warm-start works), but the recovered structure does not translate into a better MVP Sharpe.

6.3.3 Why MVP underperforms

The MVP Sharpe is dominated by the smallest eigenvalues of the inverse covariance matrix. The autoencoder's nonlinear correction perturbs these eigenvalues in-sample, and the perturbation does not generalize out of sample. PCA-based covariances (LW shrinkage, POET) preserve the eigenvalue structure that the MVP optimizer is sensitive to, so they win.

This is not a failure of the autoencoder but a mismatch between the estimand of the reconstruction problem and the estimand of the portfolio problem. The theory constrains the accuracy of the recovered common component, not the accuracy of the precision matrix. The appropriate use of autoencoder-based factor models is in tasks that depend primarily on the recovered common component—prediction, factor extraction, panel regression, matrix completion—rather than in tasks that depend on the precision matrix.

6.4 What did not work, and what we learned from it

This subsection documents two applications that we attempted and did not pursue to a successful conclusion, and one implementation issue that is worth recording. The failures are reported because they are informative about what the method can and cannot do at the sample sizes we have access to, and because they bear directly on the choice between TV-AE-G and TV-AE-L and on the future direction of the project.

6.4.1 Equity return prediction and encoder collapse

The first attempted application was monthly equity return prediction, following Gu, B. Kelly, and Xiu (2021). We pulled monthly returns for 3,521 US common stocks (share codes 10 and 11, prices above \$1, market caps above \$100 million) from Refinitiv, 2000–2024, giving 617,063 observations after dropping rows with missing return or market cap. For the first round of tests we used a 500-stock slice (top 500 permnos by mean market cap, 2010–2024) to keep training times manageable.

The first specification was TV-AE-G with no intercept term in the decoder, matching the simulation setup. The result was a total out-of-sample R^2 of 0.029 and a predictive R^2 of 0.0029. For reference, PCA on the same panel achieves 0.38 total R^2 and 0.0007 predictive R^2. The autoencoder was ten times worse than PCA on reconstruction, even though the theory says it should at least match.

The diagnosis was that the encoder collapsed to a constant. The correlation between predicted and fitted returns was 0.999998, which implies that the encoder output was essentially constant across time. Computing the standard deviation of the factor components per test year, we found values between 10^{-7} and 10^{-12}: the encoder had collapsed to outputting the per-stock mean return.

The interpretation is that the signal-to-noise ratio of monthly returns (roughly 10^{-4} per observation) combined with the high dimensionality of the encoder input (1,000 after padding and masking) made gradient descent unable to find the factor structure. The gradient was dominated by the mean, which a constant output can capture. Adding an intercept term to the decoder (so the model becomes $r_{it} = \alpha_i + \beta_i(t)'F_t$) did not fix the problem: the autoencoder still produced a constant factor, and the out-of-sample R^2 remained at 0.016.

The PCA warm-start (Section 6.2.2) fixes the collapse by initializing the encoder at the PCA loadings. With the warm-start, the autoencoder recovers the factor structure and produces an out-of-sample R^2 in the same range as PCA. But the warm-start was not enough to beat PCA on return prediction, for the reason documented next.

6.4.2 The characteristic ceiling

The second attempted fix was to add characteristics to the model, following Gu, B. Kelly, and Xiu (2021), so that the model becomes TV-AE-C: $r_{it} = \psi_i(\rho(X_t, t/T), Z_{i,t-1}, t/T) + u_{it}$. Without OSAP data, we constructed six price-based characteristics from the daily returns panel (log market cap, 12-month momentum, 1-month reversal, 12-month volatility, 12-month beta, 12-month idiosyncratic volatility), lagged by one day to avoid look-ahead.

The result was still below PCA, and the diagnosis was a hard constraint of the characteristics we could construct. In a single estimation window, we computed (i) the fraction of return variance spanned by the six characteristics and (ii) the fraction of variance captured by rank-3 and rank-5 PCA. The results:

- fraction of return variance spanned by six characteristics: 0.0192;
- fraction captured by rank-3 PCA: 0.5463;
- fraction captured by rank-5 PCA: 0.5908.

The characteristics span less than 2% of return variance. Any factor model conditioned on these characteristics can capture at most 1.92% of the variance through the characteristic channel. The remaining 98% of return variation is orthogonal to the six price-based characteristics we constructed. PCA does not suffer this limitation because its factors are constructed to maximize explained variance, not to depend on characteristics.

The ceiling is a property of the data and of the six characteristics we could construct, not of the estimator. To match Gu, B. Kelly, and Xiu (2021), we would need the 94-characteristic OSAP panel (or a comparable WRDS construction), which we did not have access to at the time. The characteristic ceiling is the binding constraint, and resolving it is future work.

6.4.3 The local estimators at $T = 250$

The local autoencoder and local PCA both fail to train at the estimation window we use for the reconstruction application ($T_{\text{est}} = 250$). The effective sample per grid point is $Th \approx 37$ observations with the Silverman bandwidth $h \approx 0.148$, which is below what a network with 8,000 parameters needs to learn anything, and below what the kernel-weighted eigenvalue decomposition needs to produce a stable estimate. Local PCA on the equity volatility panel produced $R^2 = -9.60$ (catastrophic), and on the crypto return panel gave $R^2 = 0.026$ even with an inflated bandwidth.

The local-estimator failure has a clean theoretical interpretation: the local estimator's rate $T^{-2\beta/(2\beta+K+2)}$ is asymptotically valid but requires $Th \to \infty$ with $h \to 0$, which at $T = 250$ forces $Th \lesssim 37$. The global estimator avoids this by pooling information across the full T observations, and this is why the empirical applications use TV-AE-G.

6.4.4 The weight-decay asymmetry

We mention one implementation issue because it bears directly on the empirical comparison. In an earlier run of the MVP application, the autoencoder-based covariance appeared to outperform PCA-based covariances by a substantial margin on a subset of the out-of-sample period. Investigating, we found that the autoencoder's optimizer had been given a weight-decay parameter while the PCA benchmark had not. The comparison was between a regularized model and an unregularized one, and the regularization was producing a spurious advantage.

The lesson is that autoencoder comparisons against linear benchmarks must be symmetric in regularization. This is the same lesson as in Section 5.5, where the placebo DGP detected a similar asymmetry in the simulation setup. The placebo DGP is informative precisely because it is a case where the two methods should tie: any advantage is evidence of an implementation asymmetry, not of a real improvement.

6.4.5 What these failures motivate

Three directions follow from the failure analysis, in order of feasibility.

First, a proper time-extrapolation scheme for the test-time encoder. Currently the time index is frozen at its last training value ($\tau = 1$) during the test window, which is a simplification. A ramp scheme that advances τ at the natural rate $T_{\text{te}}/T_{\text{tr}}$ improves the crypto-return out-of-sample R^2 from 0.630 to 0.642, with a bootstrap CI that marginally includes zero.

The improvement is concentrated in 2021 (the year of the strongest regime shift). A more refined scheme, such as a local-linear extrapolation of the encoder's time dependence, is left to future work.

Second, a richer characteristic set for TV-AE-C. The price-based characteristics span only 1.92% of return variance, so the conditional extension cannot add value at the current characteristic set. The OSAP or WRDS characteristic panel would remove this constraint.

Third, a longer crypto history. The crypto-return panel is thin: 50 coins from 2021 onwards, roughly 70 refit windows. The bootstrap CIs are correspondingly wide. A longer panel would sharpen the comparison and would allow a proper test of the time input's contribution over the static autoencoder.

6.5 Discussion

The empirical results, taken together, support three conclusions.

First, the autoencoder is not harmed by its additional flexibility. On panels where the true loading function is linear (equity returns, equity volatility, crypto volatility), it matches PCA within noise. This is an important robustness property: practitioners can use the autoencoder without sacrificing performance when the linear model is adequate.

Second, the autoencoder improves on PCA when the loading function is genuinely nonlinear. On the crypto return panel, the static AE gains 2 points of reconstruction R^2 over PCA, and TV-AE-G gains 3.6 points. Both gains are statistically significant or marginally significant, and the K1 sensitivity analysis confirms that the improvement peaks at a finite bottleneck width, consistent with the interpretation that the autoencoder is capturing real nonlinear structure rather than overfitting.

Third, the autoencoder's advantage is concentrated in the factor recovery problem rather than the portfolio optimization problem. This is consistent with the theory, which characterizes the autoencoder's performance for the common component $X^\star$, not for the inverse covariance matrix.

The failures documented in Section 6.4 are informative about the boundary conditions. The equity return prediction exercise fails because the signal-to-noise ratio at monthly frequency is below what gradient descent can learn from; the characteristic ceiling is a property of the price-based characteristics we could construct; and the local estimators fail because the effective sample per grid point is too small. Each of these is a constraint of the current data or of the current sample, not of the method, and each has a clear future direction.

7 Conclusion

We have developed a non-asymptotic theory and a practical estimator for nonlinear factor models with time-varying loading functions. The theory establishes the rate $T^{-2\beta/(2\beta+K+1)} + N^{-1}K_1 + \epsilon_N$ for the global time-conditioned autoencoder, characterizes the identification of the latent factors up to a time-varying invertible transformation, and extends the result to the characteristic-conditioned setting ($T^{-2\beta/(2\beta+K+P+1)} + N^{-1}K_1 + \epsilon_N$). The rate strictly dominates the rate of a local autoencoder analogue of L. Su and Wang (2017), and it reduces to the Shen and Xiu (2026) rate $T^{-2\beta/(2\beta+K)}$ when the loading function is constant in t.

The simulation evidence confirms the theory across a grid of DGPs that vary the form of the nonlinearity and the form of the time variation, and identifies the boundary case in which the static autoencoder overtakes the time-varying one at very large sample sizes. The empirical evidence is consistent with the theoretical predictions: the autoencoder's advantage materializes on panels with genuine cross-sectional nonlinearity (crypto-currency returns), and it does not materialize on panels where the linear factor model is not misspecified.

Several extensions are promising. First, the empirical exercise can be extended along three dimensions: longer crypto-currency histories, other panels with genuine nonlinearity such as option-implied volatilities, cross-country equity indices, or commodity futures, and higher-dimensional characteristic sets for TV-AE-C. Second, the panel regression with unobserved heterogeneity that Shen and Xiu (2026) develop using a double autoencoder can be extended to the time-varying setting. Third, the identification of the number of factors in nonlinear factor models remains an open problem. Fourth, the TV-AE-C estimator with characteristics has not been exercised empirically in this paper; a richer characteristic set is required to see whether the conditional extension adds value in practice.

The broader message of the paper is that the nonlinear time-varying factor model is a natural and tractable object. Deep autoencoders provide a consistent estimator with non-asymptotic guarantees and computational tractability at realistic sample sizes, and the global estimator's rate strictly dominates the local estimator's rate under smoothness in t/T. The estimator's advantage over linear methods is not universal: it is a function of whether the underlying loading function is genuinely nonlinear, and whether the sample is large enough for the nonlinearity to be identified.

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A Proof of Theorem 1

This appendix contains a self-contained proof of Theorem 1. The proof adapts Theorem 2(a) of Shen and Xiu (2026) (henceforth SX). Several approximation and concentration results are imported from the SX supplementary appendix (SX App.) and stated explicitly below.

A.1 Notation and function class

Let $\{X_{it}\}$ for $i = 1, \dots, N$ and $t = 1, \dots, T$ be the observed panel, $X_t = (X_{1t}, \dots, X_{Nt})^\top \in \mathbb{R}^N$ the cross-section, and $s_t = t/T \in (0, 1]$ the scaled time index. Let $\varphi_i^\star : \mathbb{R}^K \times [0, 1] \to \mathbb{R}$ denote the true loading function, $F_t^\star \in \mathbb{R}^K$ the latent factors, U_{it} the error, and $X_{it}^\star = \varphi_i^\star(F_t^\star, s_t)$ the common component, so $X_{it} = X_{it}^\star + U_{it}$.

Definition 1 (Time-conditioned disjoint-output autoencoder class)**.** *For $d_1, d_2, w_1, w_2 \in \mathbb{N}$ and $C, B > 0$, let $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, C, B)$ denote the set of pairs $(\rho, \{\psi_i\}_{i=1}^N)$ with $\rho \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, C, B)$ and $\psi_i \in \mathcal{F}_{K_1+1}^{1}(d_2, w_2, C, B)$ for each i, where $\mathcal{F}_{n_0}^{n_{d+1}}(d, w, C, B)$ is the ReLU network class of Shen and Xiu (2026) §2.2. By definition of that class, every member f satisfies $\|f\|_\infty \le B$. The depth parameter d_1 is an* upper bound*: the class includes networks of depth $\le d_1$, with shallower networks padded with identity layers.*

The TV-AE-G estimator is a minimizer $(\hat\rho, \{\hat\psi_i\}_{i=1}^N)$ of $L(\rho, \{\psi_i\}) := \sum_{t,i}(X_{it} - \psi_i(\rho(X_t, s_t), s_t))^2$ over $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B)$. The fitted value is $\hat X_{it}^G := \hat\psi_i(\hat\rho(X_t, s_t), s_t)$. The target quantity is $\mathcal{E}_{NT} := (NT)^{-1} \sum_{t,i}(\hat X_{it}^G - X_{it}^\star)^2$.

A.2 Imported results from SX

Proposition 3 (SX App. Proposition A1: Kohler–Langer approximation)**.** *Let $f \in \mathcal{H}^p([-a, a]^r, C)$ for $p = q + s$, $q \in \mathbb{N}_0$, $s \in (0, 1]$, $C > 0$, $a \ge 1$. Let $\zeta \ge \zeta_0(p, a, C, \|f\|_{C^q})$ sufficiently large. If $d \asymp \lceil \log_4(\zeta^{2p}) \rceil \cdot (\lceil \log_2(\max\{q, r\}+1) \rceil + 1)$ and $w \asymp 2^r \binom{r+q}{r} r^2 (q+1) \zeta^r$, then there exists a network $\hat f_{\text{wide}}$ with depth d, width w, weights bounded by ζ^{5p+5}, such that $\|\hat f_{\text{wide}} - f\|_\infty \le C' \zeta^{-2p}$ on $[-a/2, a/2]^r$, where C' depends only on $(p, a, C, \|f\|_{C^q})$.*

Lemma 1 (SX App. Lemma 5: sub-Gaussian bound)**.** *Under Assumption 1, for any fixed matrix of constants $\{C_{it}\}$ and conditional on $F^\star$, the statistic $(\sum_{t,i} C_{it} U_{it})/(\sum_{t,i} C_{it}^2)^{1/2}$ is sub-Gaussian with norm bounded by $\sigma_u > 0$, despite cross-sectional and serial dependence.*

Lemma 2 (SX App. Lemma 7)**.** *Under Assumptions 1–3, if $\log T = o(\epsilon_N^{-1})$, then with probability at least $1 - C\exp(-c\epsilon_N^{-1})$, $\|W^\star U_t\|_\infty \le 1$ for all $t \in \{1, \dots, T\}$.*

Lemma 3 (SX App. Lemma 8)**.** *Under Assumptions 1–3, with probability at least $1 - C\exp(-cT)$, $T^{-1}\sum_{t=1}^T \|W^\star U_t\|^2 \lesssim \epsilon_N$.*

Lemma 4 (SX App. Lemma 9)**.** *For $\varphi \in \mathcal{F}_{n_0}^{n_{d+1}}(d, w, T^{5\beta+5})$ and $x, y \in \mathbb{R}^{n_0}$, $|\varphi(x) - \varphi(y)| \le n_0 T^{(5\beta+5)(d+1)} w^d \|x - y\|_\infty$.*

Lemma 5 (SX App. Lemma 10)**.** *For $\mathcal{F}_{n_0}^{n_{d+1}}(d, w, T^{5\beta+5}, B)$ with weight vector $n = (n_0, \dots, n_{d+1})$ and at most S nonzero weights, there exists finite $\mathcal{F}_{n_0,\delta}^{n_{d+1}}$ with $|\mathcal{F}_{n_0,\delta}^{n_{d+1}}| \le (8\delta^{-1} C T^{(5\beta+5)(d+2)}(1 + w)^d n_0 n_{d+1} d)^{2S}$ such that for every φ in the class there exists $\overline\varphi$ in the subset with $\|\varphi - \overline\varphi\|_\infty \le \delta$ for all $\|x\|_\infty \le C$.*

Lemma 6 (Basic inequality, SX main text Eq. (13)). *For* any *element* $(\phi_i \circ \rho)$ *of* $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B$ *the minimizer* $(\hat{\rho}, \{\hat{\psi}_i\})$ *of* L *satisfies*

$$\sum_{t,i} \big(X_{it}^\star - \hat{X}_{it}^G\big)^2 \le \sum_{t,i} \big(X_{it}^\star - \phi_i(\rho(X_t, s_t), s_t)\big)^2 + 2 \sum_{t,i} U_{it} \big(\hat{X}_{it}^G - \phi_i(\rho(X_t, s_t), s_t)\big). \quad (16)$$

A.3 Auxiliary lemmas

Lemma 7 (Encoder existence). *Suppose Assumptions 1–3 hold. Let* $d_1 \asymp \log T$ *and* $w_1 \asymp T^{(K+1)/(2(2\beta+K+1))}$. *Then, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$, *there exists* $\rho^\dagger \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, T^{5\beta+5}, B)$ *satisfying*

$$\frac{1}{T} \sum_{t=1}^{T} \big\| \rho^\dagger(X_t, s_t) - (F_t^\star, \mathbf{0}_{K_1 - K}) \big\|^2 \lesssim \epsilon_N + T^{-\frac{2\beta}{2\beta+K+1}}. \quad (17)$$

Proof. Rescaling and extension. Apply the affine rescaling $\tau(z, s) := (z, (B+1)(2s-1))$, mapping $[-2B, 2B]^K \times [0, 1]$ onto $[-2B, 2B]^K \times [-(B+1), B+1]$. Let $g(z, u) := \eta^\star(z, (u/(B+1)+1)/2)$ on that image. Since g is β-Hölder with constant C_η on a box, by the Whitney extension theorem there exists $\tilde{g} \in \mathcal{H}^\beta([-2(B+1), 2(B+1)]^{K+1}, \tilde{C}_\eta)$ with $\tilde{g} = g$ on $[-2B, 2B]^K \times [-(B+1), B+1]$ and $\tilde{C}_\eta$ depending only on (β, K, B, C_η).

Approximation. Apply Proposition 3 to $\tilde{g}$ with $a = 4B$ (the domain is $[-2B, 2B]^K \times [-(B+1), B+1]$, which is contained in $[-a/2, a/2]^{K+1}$ since $B \ge 1$), $r = K+1$, $p = \beta$, $\zeta = T^{1/(2(2\beta+K+1))}$. There exists $\hat{f}_{\text{wide}}$ with depth $d_1' \asymp \log T$, width $w_1 \asymp \zeta^{K+1} = T^{(K+1)/(2(2\beta+K+1))}$, weights $\le \zeta^{5\beta+5} \le T^{5\beta+5}$, and

$$\sup_{[-(B+1), B+1]^{K+1}} |\hat{f}_{\text{wide}} - \tilde{g}|_\infty \le C_\eta' T^{-\frac{\beta}{2\beta+K+1}}. \quad (18)$$

Vector-valued $\eta^\dagger$. Apply Proposition 3 K times, one per output coordinate of $\tilde{g}$, and take the parallel composition. The total width is $Kw_1 \asymp w_1$ (K fixed, absorbed into $\asymp$). The depth $d_1' \asymp \log T$ is unchanged.

Clipping and composition. Define $\eta^\dagger(z, s) := \hat{f}_{\text{wide}}(z, (B+1)(2s-1))$, and clip the output to $[-B, B]^K$ via $C_B(y) := (\min(|y_j|, B)\operatorname{sgn}(y_j))_{j=1}^K$. Set $\rho^\dagger(x, s) := (C_B(\eta^\dagger(W^\star x, s)), \mathbf{0}_{K_1 - K})$. The input (x, s) passes through $x \mapsto W^\star x$ (two-ReLU identity) and concatenates with s via two more identity units. Total depth is at most $d_1' + 7 \asymp \log T$; pad with identity layers if needed to reach d_1. Weights bounded by $T^{5\beta+5}$: $|W_{jk}^\star| \le \|W^\star\|_2 \le (2\epsilon_N)^{1/2} \le 1$ for large N; identity and clip use weights in $\{-1, 0, 1\}$; and $\hat{f}_{\text{wide}}$ has weights $\le T^{5\beta+5}$. Thus $\rho^\dagger \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, T^{5\beta+5}, B)$.

Bounding the encoder rate. Write $z_t := W^\star X_t = W^\star \varphi^\star(F_t^\star, s_t) + W^\star U_t$, and $u_t := (B+1)(2s_t - 1)$. By Lemma 2, $z_t \in [-B-1, B+1]^K$ on a high-probability event, so $(z_t, u_t) \in [-(B+1), B+1]^{K+1}$. Since $F_t^\star \in [-B, B]^K$, $C_B(F_t^\star) = F_t^\star$, and C_B is the projection onto $[-B, B]^K$, $\|C_B(y) - F_t^\star\| \le \|y - F_t^\star\|$ for any y. Hence $\|\rho^\dagger(X_t, s_t) - (F_t^\star, \mathbf{0})\| \le \|\eta^\dagger(z_t, s_t) - F_t^\star\|$.

Decompose $\eta^\dagger(z_t, s_t) - F_t^\star$ into three terms:

$$\begin{aligned}\eta^\dagger(z_t, s_t) - F_t^\star = &\underbrace{\eta^\dagger(z_t, s_t) - \eta^\star(z_t, s_t)}_{I} \\ &+ \underbrace{\eta^\star(z_t, s_t) - \eta^\star(W^\star\varphi^\star(F_t^\star, s_t), s_t)}_{II} \\ &+ \underbrace{\eta^\star(W^\star\varphi^\star(F_t^\star, s_t), s_t) - F_t^\star}_{III}.\end{aligned}$$

Term I: $|I| \le C'_\eta T^{-\beta/(2\beta+K+1)}$ by (18). Term II: $\eta^\star$ is Lipschitz with constant KC_η, so $|II| \le KC_\eta\|W^\star U_t\|$. Term III: $|III|^2 \le 2\epsilon_N$ by Assumption 3. Squaring and applying $(a+b+c)^2 \le 3(a^2+b^2+c^2)$: $\|\rho^\dagger(X_t, s_t) - (F_t^\star, \mathbf{0})\|^2 \lesssim T^{-2\beta/(2\beta+K+1)} + \|W^\star U_t\|^2 + \epsilon_N$. Averaging over t and applying Lemma 3 gives (17). □

Lemma 8 (Decoder existence)**.** *Under Assumption 2, there exist* $\psi_i^\dagger \in \mathcal{F}^1_{K_1+1}(d_2, w_2, T^{5\beta+5}, B)$, $i = 1, \dots, N$, *with* $d_2 \asymp \log T$, $w_2 \asymp T^{(K+1)/(2(2\beta+K+1))}$, *and*

$$\sup_{(z,s)\in[-B,B]^{K_1}\times[0,1]} \left|\psi_i^\dagger(z, s) - \varphi_i^\star(z_{1:K}, s)\right| \le CT^{-\frac{\beta}{2\beta+K+1}}, \tag{19}$$

where $\varphi_i^\star$ *is extended from* $\mathcal{B}$ *to* $[-B, B]^{K_1}$ *by applying* $\varphi_i^\star$ *to* $z_{1:K}$.

Proof. Rescaling and extension. Apply $\tau(z, s) := (z, (B+1)(2s-1))$, mapping $[-B, B]^K \times [0, 1]$ onto $[-B, B]^K \times [-(B+1), B+1]$. Define $g_i(z, u) := \varphi_i^\star(z, (u/(B+1)+1)/2)$ on that image; $g_i \in \mathcal{H}^\beta([-B, B]^K \times [-(B+1), B+1], 2^\beta B)$ (the 2^β accounts for the s-rescaling in the derivative chain rule). Whitney-extend g_i to the cube $[-2(B+1), 2(B+1)]^{K+1}$, obtaining $\tilde{g}_i \in \mathcal{H}^\beta([-2(B+1), 2(B+1)]^{K+1}, \tilde{C}_{\varphi^\star})$ with $\tilde{C}_{\varphi^\star}$ depending only on (β, B).

Approximation. Apply Proposition 3 with $a = 2(B+1)$ (note: this is the decoder, whose domain is $[-B, B]^K \times [-(B+1), B+1]$, so $a/2 = B+1 \ge B$), $r = K+1$, $p = \beta$, $\zeta = T^{1/(2(2\beta+K+1))}$. There exists $\hat{f}_{i,\text{wide}}$ with depth $d'_2 \asymp \log T$, width $w_2 \asymp \zeta^{K+1} = T^{(K+1)/(2(2\beta+K+1))}$, weights $\le T^{5\beta+5}$, and $\sup_{[-(B+1),B+1]^{K+1}} |\hat{f}_{i,\text{wide}} - \tilde{g}_i|_\infty \le \tilde{C}'_{\varphi^\star} T^{-\beta/(2\beta+K+1)}$.

Clipping and padding. Since $|\varphi_i^\star| \le B$ on $\mathcal{B} \times [0, 1]$, the target $\tilde{g}_i$ satisfies $|\tilde{g}_i| \le B$ on the region where it coincides with $\varphi_i^\star$. Apply C_B outside; the composition $C_B \circ \hat{f}_{i,\text{wide}}$ satisfies $\|C_B \circ \hat{f}_{i,\text{wide}} - \tilde{g}_i\| \le \|\hat{f}_{i,\text{wide}} - \tilde{g}_i\|$ on the inner half. Define $\psi_i^\dagger(z, s) := (C_B \circ \hat{f}_{i,\text{wide}})(z_{1:K}, (B+1)(2s-1))$, zero-padded to input dimension $K_1 + 1$. This is in $\mathcal{F}^1_{K_1+1}(d_2, w_2, T^{5\beta+5}, B)$ with $d_2 := d'_2 + 2 \asymp \log T$, and inherits (19). □

Lemma 9 (Empirical process)**.** *Let* $(\rho, \{\psi_i\}) \in \mathcal{F}^{K_1}_{AE}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B)$ *be arbitrary, and set* $\tilde{X}_{it} := \psi_i(\rho(X_t, s_t), s_t)$, $s_*^2 := \sum_{t,i}(X_{it}^\star - \tilde{X}_{it})^2$. *Under Assumption 1 and the class definition, with probability at least* $1 - C\exp(-cT)$, *simultaneously for all members,*

$$\left|\sum_{t,i} U_{it}(X_{it}^\star - \tilde{X}_{it})\right| \lesssim \left(NT^{\frac{K+1}{2\beta+K+1}} + TK_1\right)^{1/2} \log^{3/2} T \cdot \left(s_* + N^{1/2}\right) + N. \tag{20}$$

Proof sketch. Grids. Let $\mathcal{S} \subset \mathbb{R}^{K_1}$ be a grid of spacing $\varepsilon_T := (K_1+1)^{-1} T^{-1-(5\beta+5)(d_2+1)} w_2^{-d_2}$ covering $[-B, B]^{K_1}$. As in SX App. Eq. (A14)–(A15), $\log|\mathcal{S}| \lesssim K_1 \log^2 T$. Apply Lemma 5 to

$\mathcal{F}^1_{K_1+1}(d_2, w_2, T^{5\beta+5}, B)$ at $\delta = T^{-1}$. Nonzero-weight count $S \asymp w_2^2 d_2 \asymp T^{(K+1)/(2\beta+K+1)} \log T$. Log-cardinality: $\log|\mathcal{F}_{2,\delta}| \lesssim S \cdot \log(T^{(5\beta+5)(d_2+2)}(1+w_2)^{d_2}) \lesssim T^{(K+1)/(2\beta+K+1)} \log^3 T$.

Deterministic coefficients. For each tuple $(\overline{\psi}_{\ell_i}, (\overline{F}_{k_1}, \ldots, \overline{F}_{k_T})) \in \mathcal{F}^N_{2,\delta} \times \mathcal{S}^T$, the coefficients $C_{it} := X^\star_{it} - \overline{\psi}_{\ell_i}(\overline{F}_{k_t}, s_t)$ are deterministic conditional on $F^\star$. By Lemma 1, $\sum_{t,i} C_{it}U_{it}/\|C\|$ is sub-Gaussian with norm σ_u. The union bound over $|\mathcal{S}|^T \cdot |\mathcal{F}_{2,\delta}|^N$ tuples and the choice $u = 2\sigma_u(T\log|\mathcal{S}| + N\log|\mathcal{F}_{2,\delta}|)^{1/2}$ give a high-probability event on which $|\sum C_{it}U_{it}|/\|C\| \le u$ simultaneously.

Discretization error. The trained $(\rho, \{\psi_i\})$ corresponds to one tuple via nearest-grid-point assignment. Pointwise, $|\tilde{X}_{it} - \overline{\psi}_{\ell_i}(\overline{F}_{k_t}, s_t)| \le 2T^{-1}$. Hence $|\sum U_{it}(X^\star_{it} - \tilde{X}_{it})| \le |\sum U_{it}(X^\star_{it} - \overline{\psi}_{\ell_i}(\overline{F}_{k_t}, s_t))| + 2T^{-1}\sum|U_{it}|$. First term: $u\|\overline{C}\|$ with $\|\overline{C}\| \le s_* + 2N^{1/2}$. Second term: $2T^{-1}\sum|U_{it}| \lesssim N$ with probability $\ge 1 - e^{-cNT}$, using $\|\Sigma\| \lesssim 1$ and Hanson–Wright. □

A.4 Proof of Theorem 1

Let $L := T^{-2\beta/(2\beta+K+1)} + N^{-1}K_1 + \epsilon_N$ and $\mathcal{E}_{NT} := (NT)^{-1}\sum_{t,i}(\hat{X}^G_{it} - X^\star_{it})^2$.

Step 1: Approximation term. Let $(\rho^\dagger, \{\psi^\dagger_i\})$ be from Lemmas 7–8. By Lemma 7, $\rho^\dagger(X_t, s_t) \in [-B, B]^{K_1}$. For each (t, i), define $T_1 := \varphi^\star_i(F^\star_t, s_t) - \varphi^\star_i(\rho^\dagger(X_t, s_t)_{1:K}, s_t)$ and $T_2 := \varphi^\star_i(\rho^\dagger(X_t, s_t)_{1:K}, s_t) - \psi^\dagger_i(\rho^\dagger(X_t, s_t), s_t)$. $|T_1| \le \sqrt{K}B\|F^\star_t - \rho^\dagger(X_t, s_t)\|$; $|T_2| \le CT^{-\beta/(2\beta+K+1)}$ (Lemma 8). Then $(NT)^{-1}\sum_{t,i}(X^\star_{it} - \psi^\dagger_i(\rho^\dagger(X_t, s_t), s_t))^2 \le C_0 L$ for some constant C_0.

Step 2: Assembly. Apply Lemma 6 with $(\phi_i \circ \rho) = (\psi^\dagger_i \circ \rho^\dagger) \in \mathcal{F}^{K_1}_{AE}$. Divide by NT, split the cross term into $\sum U_{it}(\hat{X}^G_{it} - X^\star_{it}) + \sum U_{it}(X^\star_{it} - \psi^\dagger_i(\rho^\dagger(X_t, s_t), s_t))$, and apply Lemma 9 to each. With $A_1 := (T^{-2\beta/(2\beta+K+1)} + N^{-1}K_1)^{1/2}\log^{3/2} T$ and $B_1 := (T^{-2\beta/(2\beta+K+1)-1} + N^{-1}K_1T^{-1})^{1/2}\log^{3/2} T + T^{-1}$, there exist constants $c_1, c_2 > 0$ such that $\mathcal{E}_{NT} \le C_0 L + c_1 A_1 \mathcal{E}^{1/2}_{NT} + c_1 A_1 L^{1/2} + c_2 B_1$.

Step 3: Magnitudes. $A_1^2 \le L\log^3 T$. Since $2\beta/(2\beta+K+1) < 1$, $T^{-2\beta/(2\beta+K+1)} > T^{-1}$, so $L \ge T^{-1}$; hence $(L/T)^{1/2} \le L$, giving $B_1 \lesssim L\log^{3/2} T$. Also $A_1 L^{1/2} \lesssim L\log^{3/2} T$.

Step 4: Closing. Set $v = \mathcal{E}^{1/2}_{NT}$, $c = c_1 A_1$. From $v^2 \le C_0 L + 2cv + c_1 A_1 L^{1/2} + c_2 B_1$, completing the square gives $v \le 2c + (C_0 L + c_1 A_1 L^{1/2} + c_2 B_1)^{1/2}$, hence $\mathcal{E}_{NT} \le 8c^2 + 2C_0 L + 2c_1 A_1 L^{1/2} + 2c_2 B_1 \lesssim L\log^3 T$.

B Proof of Proposition 1

The proof is Theorem 1 with the decoder dimension changed from $K_1 + 1$ to $K_1 + P + 1$, and the loading-function dimension from $K + 1$ to $K + P + 1$. The encoder is unchanged: it still outputs K_1 from (X_t, s_t), and the characteristics enter only through the decoder. We indicate the three places where the argument changes.

Decoder existence. Replace Lemma 8 with the following. Under the additional assumption $\varphi^\star_i \in \mathcal{H}^\beta(\mathcal{B} \times [-B_Z, B_Z]^P \times [0, 1], B)$, apply Proposition 3 with $r = K + P + 1$ (the decoder's input is (bottleneck, Z, s), of dimension $K_1 + P + 1$, and the target depends on (F, Z, s) of dimension $K + P + 1$) and $\zeta = T^{1/(2(2\beta+K+P+1))}$. The resulting network has width $w_2 \asymp T^{(K+P+1)/(2(2\beta+K+P+1))}$ and sup error $CT^{-\beta/(2\beta+K+P+1)}$.

Empirical process. Replace Lemma 9 with the same statement where $T^{(K+1)/(2\beta+K+1)}$ is replaced by $T^{(K+P+1)/(2\beta+K+P+1)}$. The proof is identical: the decoder class now has input

dimension K_1+P+1 and width $w_2 \asymp T^{(K+P+1)/(2(2\beta+K+P+1))}$, so the entropy bound $\log|\mathcal{F}_{2,\delta}|$ picks up $T^{(K+P+1)/(2\beta+K+P+1)}\log^3 T$ instead of $T^{(K+1)/(2\beta+K+1)}\log^3 T$. The factor grid $\mathcal{S}$ is unchanged because the bottleneck output is still K_1-dim.

Assembly. The basic inequality and Step 2–Step 4 of the proof of Theorem 1 proceed verbatim, with $L := T^{-2\beta/(2\beta+K+P+1)} + N^{-1}K_1 + \epsilon_N$ and the decoder sup error $T^{-\beta/(2\beta+K+P+1)}$. The final rate is (11). □

C Proof sketch for Proposition 2

The argument adapts Shen and Xiu (2026)'s proof of Theorem 3 with $K \to K+1$ and the encoder output $\hat{\rho}(X_t, s_t) \in \mathbb{R}^{K_1}$ in place of their $\hat{\rho}(X_t)$. The structural steps are:

Reduction to the linear case. The analogue of SX App. Lemma 14 (their "typical arguments" lemma) is needed with the encoder evaluated at (X_t, s_t) and the target at $F_t^\star$. This lemma bounds the difference between the in-sample loss and the out-of-sample loss using Bernstein's inequality for the networks in $\mathcal{F}_{N+1}^{K_1}$ and the grid $\mathcal{S} \subset \mathbb{R}^{K_1}$ of the empirical-process lemma. The bound picks up the entropy terms $\log|\mathcal{F}_{N+1}^{K_1}|$ and $\log|\mathcal{S}|$ from Lemma 9, giving a rate of $(T^{-2\beta/(2\beta+K+1)} + T^{-1/2}L_N^{1/2})\log^3 T$.

Substitution of the encoder. With μ_t defined as the map that sends the encoder output to the factor via the inverse of $W^\star\varphi^\star(\cdot, s_t)$ (extended to a map on $\mathbb{R}^{K_1}$ by zero padding), the triangle inequality yields

$$\frac{1}{T}\sum_{t=1}^{T}\|\mu_t(\hat{\rho}(X_t, s_t)) - F_t^\star\|^2 \lesssim \epsilon_N K_1 + \frac{1}{T}\sum_{t=1}^{T}\|\hat{\rho}(X_t, s_t) - (F_t^\star, \mathbf{0})\|^2 \cdot (1 + \|\eta^\star\|_\infty^2).$$

The first term is the encoder approximation error from Assumption 3 times the bottleneck dimension. The second term picks up (i) the in-sample encoder rate $T^{-2\beta/(2\beta+K+1)} + \epsilon_N$ from Lemma 7 multiplied by $N\epsilon_N$ (the noise in $\hat{\rho}$ compounds through the decoder in the empirical-process bound), and (ii) a second-order term $N\epsilon_N^2$ from the quadratic approximation of $\eta^\star$ around the noiseless factor. Combining gives (12). □

D The local autoencoder: comparison

The direct nonlinear analogue of L. Su and Wang (2017) replaces the global loss (7) with a kernel-weighted loss at each of $m \asymp 1/h$ grid points. Let $\{x_1, \ldots, x_m\} \subset (0,1)$ be a grid with uniform spacing $O(h)$, and let $\{K_h\}$ be the Epanechnikov kernel with bandwidth h. At each grid point x_j, the local autoencoder minimizes

$$(\hat{\rho}_j, \{\hat{\psi}_{i,j}\}_{i=1}^N) = \arg\min \sum_{t=1}^{T} K_h\left(\frac{t/T - x_j}{h}\right)\sum_{i=1}^{N}\left(X_{it} - \psi_i(\rho(X_t), x_j)\right)^2. \tag{21}$$

The reconstruction at time t averages across grid points with the same kernel weights.

Proposition 4 (In-sample rate for TV-AE-L)**.** *Suppose the conditions of Theorem 1 hold and let* $h \asymp T^{-1/(2\beta+K+2)}$*. Then, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{NT}\sum_{t,i}\big(\hat{X}_{it}^{L} - X_{it}^{\star}\big)^2 \;\lesssim\; \big(T^{-\frac{2\beta}{2\beta+K+2}} + N^{-1}K_1 + \epsilon_N\big)\log^4 T, \tag{22}$$

where $\hat{X}_{it}^{L}$ *is the local autoencoder reconstruction.*

Proof sketch. Apply Theorem 1 at each grid point with the kernel weight as the effective sample. The local fit at x_j uses $\asymp Th$ observations, and the loading function $\varphi_i^{\star}(F, s)$ evaluated at $s = x_j$ is still $(K+1)$-dimensional. The nonparametric error at each grid point is therefore $(Th)^{-2\beta/(2\beta+K+1)}$, and the smoothing bias in the s dimension is $h^{2\beta}$. Balancing the two gives $(Th)^{-2\beta/(2\beta+K+1)} \asymp h^{2\beta}$, which yields $h \asymp T^{-1/(2\beta+K+2)}$ and the rate $T^{-2\beta/(2\beta+K+2)}$. The remaining terms $N^{-1}K_1$ and ϵ_N are unchanged from Theorem 1. Summing over the grid and applying the kernel weighting introduces a log factor, giving the stated $\log^4 T$. □

The rate $T^{-2\beta/(2\beta+K+2)}$ is strictly slower than the TV-AE-G rate $T^{-2\beta/(2\beta+K+1)}$ because the exponent in the denominator is $2\beta+K+2$ rather than $2\beta+K+1$. The gap is the kernel-smoothing penalty of local estimation, and it is the reason the empirical work of Section 6 uses TV-AE-G.